

REVIEW PAPER

Combining Molecular Biomarkers with Low-Dose CT for Early Lung Cancer Detection

Shrinath Bathey¹, Catherine Chan², Jayani Guntumadugu³, Stella Kyae Sin Win Htet⁴, Amanjot Parmar⁵, Janelle Razunguzwa⁶, Vivian Zhao⁷

¹ Plano West Senior High School, Plano, TX, USA

² Ridge High School, Basking Ridge, NJ, USA

³ Independence High School, Frisco, TX, USA

⁴ Harrow International School Bangkok, Bangkok, Thailand

⁵ Middlesex High School, Middlesex, NJ, USA

⁶ Brunswick High School, Brunswick, MD, USA

⁷ California High School, San Ramon, CA, USA

Abstract

This systematic review analyzes past trials from the last 16 years in order to confirm whether the combination of LDCT with molecular biomarkers provides a more accurate and efficient diagnosis for lung cancer than LDCT alone. A total of 42 clinical trials and patient studies combining LDCT and biomarker trackage were sourced from PubMed using keywords like “LDCT”, “lung cancer diagnosis” From the selected studies, we extracted diagnostic performance metrics, specifically sensitivity, specificity, and area under the curve (AUC) of all the selected studies and calculated the unweighted mean per molecular biomarker. The change in sensitivity for circulating tumor cells (CTCs), protein biomarkers, circulating tumor DNA (ctDNA), DNA methylation markers, autoantibodies are as follows: 17.63, 30.96, 5.18, 32.48, and -18.54. The change in specificity for circulating tumor cells (CTCs), protein biomarkers, circulating tumor DNA (ctDNA), DNA methylation markers, autoantibodies are as follows: 26.2, 10.54, 35.29, 10.74, and 27.05. The change in AUC for circulating tumor cells (CTCs), protein biomarkers, circulating tumor DNA (ctDNA), DNA methylation markers, autoantibodies are as follows: -0.042, -0.22, -0.141, 0.027, -0.089. Taking these calculations into consideration, we’ve concluded the order from most effective molecular biomarker and LDCT pairing to least to be DNA methylation markers, CTCs, ctDNA, protein Biomarkers, to autoantibodies.

Keywords: Low-dose CT screening, Lung Cancer, Circulating Tumor Cells, DNA Methylation Markers, Protein Biomarkers, Circulating Tumor DNA, Autoantibodies

I. Introduction

Lung cancer has historically lacked research efforts compared to other cancers and also is difficult to detect because symptoms heavily overlap with other common lung conditions. Low-dose CT (LDCT) screenings, a common lung cancer detection method, has several drawbacks: an overdiagnosis of slow-growing cancers and unnecessary follow-up imaging and invasive biopsies being some of them. To improve the specificity of the LDCT screening, combining imaging with molecular biomarkers can help improve diagnostic accuracy. We looked at the combination of 5 different molecular biomarkers with the use of LDCT screenings. Introduced into the body via cancer, CTCs are cancer cells that break away from the primary tumor and enter the bloodstream. Testing these cells via a liquid biopsy can offer the possibility of assessing the tumor's characteristics, evolution, and resistance/response to treatments. Next, ProGRP is extremely specific to tumors of neuroendocrine origin, and when small cell lung cancer (SCLC) develops, ProGRP levels rise from baseline to highly elevated amounts. ProGRP, as the main protein biomarker for differentiating SCLC from non-small cell lung cancer, can improve LDCT screening by acting as a complementary biomarker to rule out false positives. Circulating tumor DNA (ctDNA) also provides an early way to detect lung cancer tumors as growing tumors often shed tiny DNA fragments in the blood before symptoms even develop, which is a major issue of why diagnosis fails to address pathogenesis in time. Similarly, genetic information from DNA methylation markers have unique tags that allow for tumor identification and can be tracked before symptoms appear. Finally, tumor associated antigens (TAAs) on autoantibodies are an immune response that can indicate bold responses to already existent growth and can identify high-risk patients.

II. Methodology

A total of 54 initial articles were identified from the PubMed database using keywords related to "LDCT", "lung cancer", "screening", "diagnosis", and various biomarker categories ("circulating tumor cells", "proteomic", "circulating tumor DNA", "DNA methylation", and "autoantibodies"). The first screening eliminated all grey literature (literature reviews, editorials, etc), articles published prior to 2010, studies with irrelevant data, and non-English articles and excluded articles were replaced with a sufficient study. Then, a final 2nd screening was performed, removing any primary literature without rigorous methodologies or biased trial designs. A total of 42 final documents were included in the final systematic review, including mainly cohort, diagnostic, and validation studies.

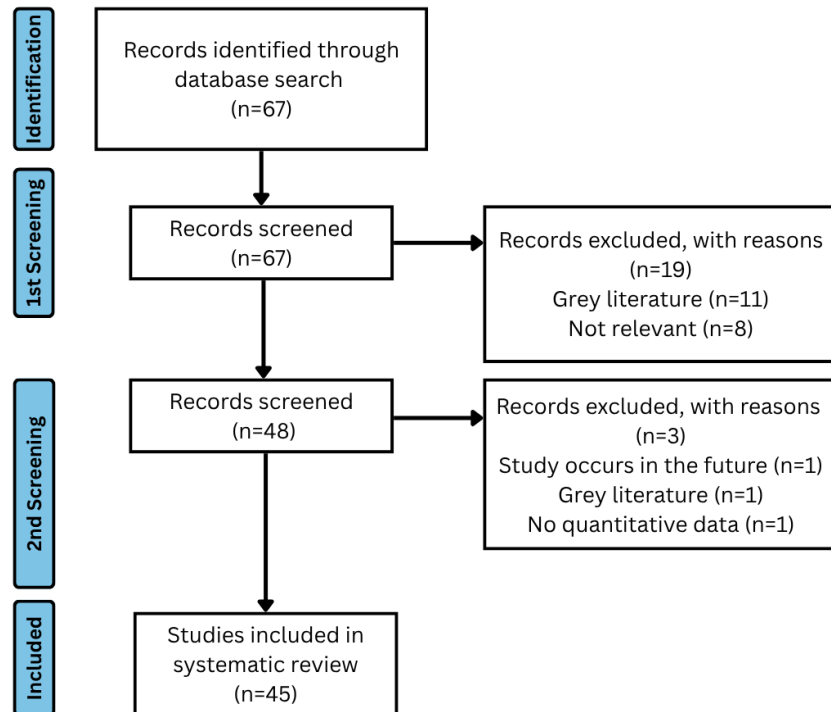
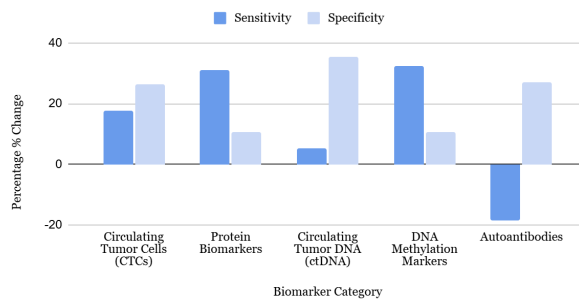


Figure 1. PRISMA Flow Diagram

III. Results

Average Difference in Sensitivity and Specificity between LDCT and Molecular Biomarker Combination and LDCT alone



Average Difference in AUC between LDCT and molecular biomarker combination and LDCT alone

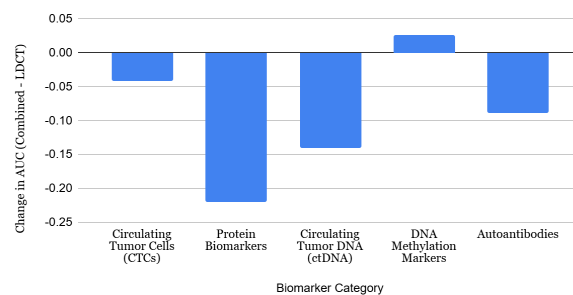


Figure 2. Diagnostic performance metrics were averaged across studies and baseline data of LDCT-only trials were subtracted from the averages of LDCT studies with molecular biomarkers to compare efficacy in lung cancer diagnosis.

1. DNA Methylation Markers

Eight studies published in 2016 to 2025 were assessed to determine the value of epigenetic and circulating genetic biomarkers for the diagnosis, prognosis, and prediction of lung cancer response to treatment. The general aim of these studies was to evaluate the efficiency of detecting DNA methylation patterns and circulating tumor DNA as screening tests for lung

cancer compared to standard-of-care imaging, risk prediction questionnaires, and cytopathology. The objectives of the studies included the identification of high-risk patients who had not yet presented symptoms, differentiation between cancerous and benign lung nodules at the early stage of the disease, and assessment of the response to treatment.

Among the eight studies, the designs included prospective cohort, observational cohort, clinical diagnostic trials, and diagnostic cohort studies. The years of the included research data ranged from before 2016 to 2024 with a range of 50 patients in small prospective cohorts and 1480 patients in large multinational pooled analyses. In the studies researchers assessed the association in ever-smoking populations, patients with suspected pulmonary nodules, patients diagnosed with early-stage IA or IB non-small cell lung cancer (NSCLC), those with advanced-stage III or IV disease who were subjected to immunotherapy, and those with benign pulmonary disease or no cancer. The researchers conducted the trials in Australia, Sweden, Italy, Norway, China, Spain, and the USA. The range of cancer stages included in the findings ranged from pre-invasive lesions such as adenocarcinoma in situ and minimally invasive adenocarcinoma to early-stage I NSCLC and advanced-stage III or IV.

The control groups consisted of people who didn't have cancer and were similar in age and smoking habits to patients with cancerous lung growths, tissue samples that were not cancerous, or blood levels before any treatment. The factors they looked to for diagnosis included tests for many genes that had been turned off like CDO1, SOX17, TAC1, HOXA7, BCAT1, TRIM58 ZNF177, SHOX2 and RASSF1A. They also looked at the overall pattern of a certain chemical change in the genes' special risk scores from blood tests before people got sick and the genetic changes in cancer DNA that was floating in the blood, and took samples from parts of the body like tissue, blood, plasma, sputum, liquid from the lungs and other fluids. After, they used laboratory techniques like a test that can find specific chemical changes, a machine that can measure these changes a way to look at the genes on tiny beads a special microscope that can see in 3D, a test that can find out the exact order of the genes and a way to look at the genetic code of specific parts of the DNA. These tests were used by themselves or with other information like how likely someone was to get cancer, what the cancer looked like on an X-ray or if they had certain genetic changes, like EGFR, to see if these cancer tests could help find cancer.

Single gene methylation analyses showed little diagnostic value, correctly identifying methylated targets only 41–76% of the time in blood and 63–86% of the time in sputum. Multi-marker panels, on the other hand, were much more effective at detecting cancerous sites: three-gene blood tests had a sensitivity of 90% and a false-positive rate of 29% (or 91% sensitivity and 21% false-positives when including patient demographics), sputum multi-marker panels had sensitivities of 84.6–100%, and single-cell nuclear topology had a 100% sensitivity and 26.1% specificity. Multi-marker analysis of tissue samples and combined methylation and EGFR analysis had 60.9% sensitivity and 80% specificity for identifying invasive nodules, but dropped drastically in accuracy for detecting sub-centimeter tumors in plasma (64%) and pre-invasive lesions in tissue (16.1%). Adding pre-diagnostic blood methylation to the

PLCOM2012 risk model had an area under the curve of 0.73, which showed no statistically significant improvement on the diagnostic capability of the questionnaire alone.

Multi-gene panels, including CDO1 and SOX17, are superior to single biomarker tests and cytology in combination with clinical and/or radiographic assessment for early lung cancer detection in various biofluids (Chen et al., 2020; Díaz-Lagares et al., 2016; Hulbert et al., 2017). Methylation in combination with EGFR status allows to effectively classify subtypes of small pulmonary nodules (Ji et al., 2023). However, the detection of lung cancer by methylation is less effective for small-sized and non-invasive tumors (Chen et al., 2020; Zhao et al., 2024). Nuclear topology from sputum can be used for early lung cancer detection, whereas specificities are reduced when compared to benign diseases (Soukiasian et al., 2023). Although pre-diagnostic blood methylation is largely reflective of accumulated tobacco smoke exposure, it rarely represents the presence of tumor DNA (Onwuka et al., 2024). Serial measurement of ctDNA clearance from blood can inform immunotherapy response in patients with advanced NSCLC (Gelmini et al., 2025), so blood methylation analysis reflects environmental exposures, while multi-gene airway, plasma, and tissue panels are used for the detection of early cancers, monitoring of invasion, and response to therapy.

2. Circulating Tumor Cells (CTCs)

Eight studies published between the period of 2017 to 2025 were evaluated to analyze the value of circulating tumor cell (CTCs) biomarkers in the early detection and classification of lung cancer and pulmonary nodules. The general objective was to determine the effectiveness of collecting circulating tumor cells and combining them with additional biomarkers for improving lung cancer screening. The aims of the studies included differentiating malignant from benign pulmonary nodules, identifying high-risk patients, and risk prediction.

Researchers utilized study designs that included cross-sectional cohort, prospective cohort, diagnostic cohort, and validation cohort studies. Data for these studies were collected from 2007 to 2020. The samples for them ranged from 118 individuals to upwards of 8,313 in larger, multicohort studies spanning multiple hospitals. The populations included high-risk individuals that were heavy smokers, patients diagnosed with lung cancer, individuals that had or likely had pulmonary disease, and an array of patients from Chinese-based hospitals with trials conducted in China, France, and the United States. The stages of lung cancer detected within the studies included both early and severe stages of NSCLC in patients.

Control groups mainly consisted of individuals that were healthy, usually adult volunteers or had benign pulmonary nodules; occasionally there were multiple groups that all received the same treatment. Evaluating CTCs, researchers usually collected peripheral blood in patients that had previously gone through LDCT screenings. Afterwards, the studies diverged in their ways. Xu et al. (2023) and Marquette et al. (2017) utilized the isolation by size of epithelial tumor cells (ISET) technique along for CTC counts. Ma et al. (2022) used negative enrichment-fluorescence in-situ hybridization (NE-FISH) for CTC detection with serum tumor marker panels (CEA, CA125, CYFRA21-1, SCC), while Katz et al. (2020) identified used antigen-independent 4-color

FISH. He et al. (2017) and Jin et al. (2025) developed newer technologies that split them from the others—inserting a CellCollector wire (anti-EPCAM-coated) into the arm vein directly to identify the subsequent CTCs and placing their Labyrinth microfluidic chip directly in 5 mL blood. Liu et al. (2021) utilized a folate receptor (FR)-positive CTC detection kit with serum tumor markers (CEA, CA125, CA724, CYFRA21-1, NSE). Many studies delved into additional tools and imaging to refine results.

The studies utilized a variety of diagnostic performances. In the study investigating the CellCollector wire, it detailed how CTCs identified 73.3% of individuals that had lung cancer (He et al, 2017). It achieved a sensitivity of 52.6% and specificity of 80.0% in differentiating benign from malignant pulmonary nodules with LDCT screening, with positive and negative predictive values of 91.1% and 30.3% along with an AUC value of 0.651 (Xu et al., 2023). Marquette et al. (2017) and their peripheral blood CTC study found when considering both the CNHC-malignant and CNHC-uncertain as positive, the CTC detection specificity was 96.2% , the negative predictive value was 97.6%, and the positive predictive value was 18.42%. In another study, for LDCT screening Jin et al. (2025) achieved a sensitivity of 84.4%, specificity of 82.4%, AUC and of 0.84. Overall, sensitivity was 84.4%, specificity was 88.2%, AUC was 0.90. In the two studies that utilized fluorescence in situ hybridization, the using CTC counts and tumor marker levels had a sensitivity of 80.12% and AUC of 0.853 for the differential diagnosis of PNs (Ma et al, 2022), and in the other study the identification of CTCs using FISH probes detected lung cancer cases with 94.2% accuracy, 89% sensitivity, and 100% specificity compared with biopsy (Katz et al., 2020). Another study with CTC cluster found them in 41.4% of all NSCLC patients tested (Manjunath et al., 2019), while the study utilizing a folate receptor (FR)-positive CTC detection kit found CTCs in diagnosing malignant solitary pulmonary nodules (SPNs) has a sensitivity of 89.1%, specificity of 92.3%, and AUC of 0.907 (Liu et al., 2021).

In general, combining biomarkers such as CTCs with LDCT screening can significantly impact the detection of early stage lung cancer. Many studies indicate that there was success and the potential for CTCs to play an ancillary role for differentiating malignant pulmonary nodules from benign ones (Xu et al., 2023; Ma et al., 2022; Katz et al., 2020; Manjunath et al., 2019; Lie et al., 2021). FISH methods like CTC assays help support LDCTs by providing a noninvasive tool to conduct liquid biopsies, having a high sensitivity and specificity when compared to other methods (Katz et al., 2020). Additionally, CTC clusters may serve as a more specific candidate for NSCLC detection than single CTCs (Manjunath et al., 2019), proving as a superior diagnostic tool than other biomarkers due to tumor heterogeneity (Liu et al., 2021). Furthermore, studies combining new, experimental platforms with LDCTs provide evidence for their effectiveness and potential as opposed to prior EpCAM or ISET testing methods (Xu et al., 2023; He et al., 2017). In all, researchers have found very promising results from CTCs that present viable options when compared to other biomarkers for early-stage lung cancer detection and risk predicting, though there may be a need to conduct routine LDCT screening to realize their full effectiveness (He et al. 2017; Marquette et al., 2017).

3. Circulating Tumor DNA (ctDNA)

Ten studies on ctDNA biomarkers published between 2020 and 2026 were thoroughly analyzed to investigate their role in the diagnosis, early detection and relapse of non-small cell lung cancer (NSCLC). The general objective of the conducted research was to determine the efficacy of ctDNA biomarkers in the diagnosis and screening of lung cancer and minimal residual disease (MRD). The aims of these studies was to evaluate different clinical applications of ctDNA including early detection of NSCLC, minimal residual disease (MRD) detection, recurrence prediction and risk stratification (Gale et al., 2022; Tian et al., 2024; Oh et al., 2024; Tan et al., 2024; Jiang et al., 2020; Huang et al., 2024; Walls et al., 2020; Cha et al., 2026; Wang et al., 2023; Black et al., 2025).

In terms of study characteristics, our studies investigated the early detection potential of ctDNA (Jiang et al., 2020; Cha et al., 2026; Wang et al., 2023; Black et al., 2025). The remaining studies focused on postoperative MRD and recurrence rates (Gale et al., 2022; Tian et al., 2024; Oh et al., 2024; Tan et al., 2024). Huang et al. specifically examined the concordance between ctDNA-based and tissue-based next-generation sequencing (NGS) in patients with advanced NSCLC (Huang et al., 2024). The study designs involved prospective cohort studies, biomarker validation, diagnostic case-control and cross-sectional studies. Sample sizes also ranged from 1323 plasma samples (Wang et al., 2023) to only 4 patients (Walls et al., 2020). They mostly examined stages I-III in terms of disease, although a few included stage IV of NSCLC.

ctDNA usually involves the collection of peripheral blood, then extraction from the plasma and molecular analysis of the DNA. Several studies used tumor-informed approaches, where tumor tissue was sequenced first to identify patient-specific mutations, like patient-specific RaDaR assays, WES and personalized NGS assays (Gale et al., 2022; Tan et al., 2024). There were also broader targeted sequencing approaches like a 560-gene panel to detect mutations and a 425-gene panel to compare NGS assays (Jiang et al., 2020; Huang et al., 2024). Diagnostics were collected at different time periods and control groups were also used such as healthy individuals, benign nodules and patients without COPD (Jiang et al., 2020; Wang et al., 2023; Cha et al., 2026).

Multiple diagnostic performances were used. In the study investigating RNF213 mutations, ctDNA helped provide 100% specificity in identifying this mutation in all samples (Jiang et al., 2020). The LunaCAM-D assay demonstrated 96.7%, a PPV of 84.2% and an NPV of 76.1% (Wang et al., 2023). It was also found that adding molecular and genomic information improved the ctDNA model sensitivity 52.1% to 63.9%, and the area under the curve (AUC) from 0.620 to 0.729 (Cha et al., 2026). NeXT Personal study also granted high detection rates: 81% in NSCLC patients with lung adenocarcinomas and 100% in NSCLC patients lacking it (Black et al., 2025). For postoperative applications, ctDNA offered a high diagnostic performance, with Gale et al. reporting 86.7% sensitivity, 92.9% PPV and 87.5% NPV for recurrence detection (Gale et al., 2022).

Overall, ctDNA has strong potential as a minimally invasive biomarker for high-risk patients with lung cancer. However, ctDNA may currently be the most effective in detecting postoperative MRD and recurrence detection, since higher levels of ctDNA are associated with a higher risk of relapse (Gale et al., 2022; Tian et al., 2024; Oh et al., 2024; Tan et al., 2024). The evidence for primary diagnosis and early detection was also promising, especially newer and more sensitive approaches like LunaCAM and NeXT Personal (Wang et al., 2023; Black et al., 2025). However, limitations include poor generalisability due to small sample sizes, single-centre recruitment, retrospective study designs and inconsistent sampling periods (Tian et al., 2024; Oh et al., 2024; Jiang et al., 2020; Cha et al., 2026). More importantly, there is still a lack of studies precisely comparing ctDNA with LDCT. Further research is still needed to determine whether it is superior to imaging techniques such as LDCT.

4. Protein Biomarkers (ProGRP)

Eight studies published from 2014 to 2023 evaluated the use of circulating blood-based proteins to predict the risk of developing lung cancer and assess classification of indeterminate pulmonary nodules. The goal was to identify malignant versus benign nodule status, detect lung cancer from stages I to IV, and improve risk assessment for high-risk smokers beyond LDCT and standard clinical models. The majority of studies incorporated protein test results with imaging features and epidemiologic risk factors to develop screening and nodule management strategies.

The eight research studies utilized a mix of case-control designs with independent validation samples and multicenter observational trials. The sample sizes in the studies had a wide scale with sample sizes ranging from 200 patients to more than 1,800 people in multicancer tests, along with nearly 1,300 blood samples in U.S. screening trials. The participants were a diverse group made up of high risk current and former smokers, people with benign and malignant lung lesions, healthy donors, and patients with CT-detected nodules. Most of the research was focused on lung cancer cases Stages I to II NSCLC specifically looking at early stage tumors that could be surgically removed and suitable for LDCT screening.

To analyze the multiple circulating protein samples from plasma or serum, the researchers used MRM mass spectrometry and LC-PRM-MS. The protein biomarkers that were commonly included were Pro-SFTPb, CA125, CYFRA 21-1, CEA, CEACAM5, ERBB2, MET, and CXCL1. There were multiple multi-analyte assays that were used to test for circulating tumor DNA mutations. Logistic regression and supervised machine learning were used to calculate malignancy probabilities. Multiple biomarker models integrated CT nodule size, Lung-RADS LDCT findings, the Mayo Clinic calculator, or PLCOm2012 for more accurate screening eligibility.

The diagnostic accuracies of the biomarkers depended on the population and specific application. The 13-protein classifier reached a negative predictive value ranging from 90% to 94% and a sensitivity of approximately 92%, but a specificity of only 44%. This made it useful for ruling out surveillance, but still required additional follow-up on some patients (Li et al., 2014). The six-protein LC-PRM-MS panel had an AUC near 0.999 with sensitivity and

specificity near 99%. Healthier controls were limited to typical LDCT populations (Uribarri et al., 2020). The four-protein serum panels combined with CT nodule size had AUCs of 0.95 to 0.96, sensitivity near 82%, and specificity above 85%. Its addition to PLCOm2012 led to an AUC increase from approximately 0.70 to the mid-0.80s (Fahrman et al., 2022; Lu et al., 2021; Ostrin et al., 2022; Toumazis et al., 2023).

Overall, using protein biomarkers has improved the ability to assess lung cancer risk when combined with LDCT and CT results. Classifiers with high negative predictive values can decrease the frequency of unneeded biopsies and intensive CT scans for patients with likely benign nodules. In the future, protein panels paired with cell-free DNA could be beneficial for both early detection and risk prediction. However, before routine implementation of protein biomarkers, larger and more diverse cohorts and standardized clinical trials are necessary (Cohen et al., 2018; Fahrman et al., 2022; Li et al., 2014; Lu et al., 2021; Ostrin et al., 2022; Toumazis et al., 2021, 2023; Uribarri et al., 2020).

5. Autoantibodies (AABs)

8 articles from 2017-2026 were evaluated for diagnostic performance of AABs in lung cancer diagnosis and pulmonary nodule evaluation, consisting of retrospective diagnostic accuracy studies, retrospective validation cohorts, observational studies, and retrospective pooled cohort studies of lung nodules. Each study had a population size of 178-1164 participants and all used AAB utility in either lung cancer diagnosis or pulmonary nodule evaluation. Most populations included patients with either CT-detected or undefined pulmonary nodules, and were conducted in the US, Canada, Germany, and China.

Methods included the seven-AAB EarlyCDT-Lung test, 7-AAB test panels, and 4-marker tumor-derived AAB panel. Current study evaluated if blood-based AABs could differentiate benign from malignant pulmonary nodules, and if diagnostic accuracy could be increased with blood-based tests when combined with CT imagery, demographics, and known lung cancer risk factors. Sample sizes varied from 170-1200 patients who either had pulmonary nodules, benign tumors, or served as healthy controls (Gonzalez Maldonado et al., 2021; Guo et al., 2024; Huo et al., 2020; Lastwika et al., 2019; Liu et al., 2024; Long et al., 2025; Massion et al., 2017; Pitcher et al., 2026;). Types of cancer included non-small cell lung carcinoma (NSCLC), small cell lung carcinoma (SCLC), carcinoid tumors, adenocarcinoma, and squamous cell carcinoma with most studies investigating early stage non-small cell lung carcinoma on CT with AAB profiling.

Most AAB tests identified antibodies through ELISA with serum or plasma, while others tested seven-AAB signatures. The EarlyCDT-Lung assay tested for AABs against seven different lung cancer antigens (Massion et al., 2017; Long et al., 2025; Pitcher et al., 2026; Gonzalez Maldonado et al., 2021), with other tests examining panels of GAGE7, CAGE, MAGEA1, SOX2, GBU4-5, PGP9.5, P53, and other cancer antigens (Huo et al., 2020; Guo et al., 2024). Liu et al. (2024) determined that using TAABs along with CT and demographic variables helped determine cancer with logistic regression. Lastwika et al. (2019) created a four-marker AAB panel of FCGR2A, EPB41L3, LINGO1 IgG and S100A7L2 IgM based on tumor-derived

antibody signatures. Indeterminate nodules were verified by the stability or resolution of imaging or by clinical follow-up with six-month follow-up in the studies of Massion et al. (2017) and Huo et al. (2020).

The diagnostic performance varied markedly with different AAb assays and patient groups. The majority of the studies utilizing the 7-AAb EarlyCDT-Lung-type panels reported high specificity but low sensitivity. The results of Long et al. (2025) reported 16% sensitivity, 90% specificity, 66% PPV; while the results of Pitcher et al. (2026) reported 16% sensitivity, 91% specificity, 50% PPV. The results of Massion et al. (2017) reported 13% sensitivity, 97% specificity, 50% PPV for the AAb test alone. In combination with a Mayo Clinic risk-factor cutoff of 30%, the specificity increased to 99%, sensitivity increased to 25%, and PPV increased to 91%.

Other AAb assays had other various results. Guo et al. (2024) had 84.05% sensitivity, 91.85% specificity and an AUC of 0.8870, which increased to 0.9265 with the SSA-XGBoost model. Huo et al. (2020) had 45.5% sensitivity, 83.3% specificity and an AUC of 0.660 which increased to 55.4% sensitivity and an AUC of 0.719 with an alternative interpretation. Liu et al. (2024) had 44.76% sensitivity and 76.71% specificity for TAABs alone, increasing to 61.5% sensitivity and 87.1% specificity when TAABs were combined with CT scan and demographics. Lastwika et al. (2019) had an AUC of 0.737 with 33.3% sensitivity at 90% specificity, but sensitivity increased to 91.7% at 57.1% specificity in an observed set of patients with indeterminate 8–20 mm nodules. Across the included studies, AAb testing was found to be useful as an adjunct to the risk assessment for lung cancer and assessment of pulmonary nodules but not as a single screening or diagnostic modality. The EarlyCDT-Lung and similar seven-AAb panels had consistently high specificity but low sensitivity, which identified some patients as being at risk but lacked the ability to reliably exclude lung cancer in negative tests. The combination of AAb testing with CT scanning, demographics and established risk factors showed an improvement in diagnostic accuracy as demonstrated by Liu et al. (2024) and Massion et al. (2017). Guo et al. (2024) demonstrated high sensitivity in a highly selected patient group but AAb performance may be influenced by patient selection and disease specific factors.

The high heterogeneity of retrospective design, small number of cancer positive patients, variety of different analytical methods, and limited follow up periods are subject to bias. and Further research and external validation is needed to confirm the success of autoantibodies for treatment.

IV. Conclusion

1. DNA Methylation Markers

The clinical utility of DNA methylation markers stems from their ability to improve early lung cancer diagnosis, prognosis, and facilitate non-invasive triage. Epigenetic changes take place early during tumorigenesis and can, therefore, be used for the development of tissue, plasma, or airway fluid (sputum and bronchoalveolar lavage)-based methylation panels. The diagnostic accuracy of single-gene assays remains modest with the reported sensitivity ranging

from 41 to 76% for plasma tests. In contrast, the sensitivity of multi-gene assays (CDO1, SOX17, TAC1, HOXA7, BCAT1, TRIM58, ZNF177, SHOX2, and RASSF1A) is substantially higher, with the accuracy of blood-based tests ranging from 90 to 91%. The sensitivity of sputum-based tests can reach up to 100%, while methylation analysis of tissue samples combined with the assessment of driver mutation status (EGFR) can help to differentiate between invasive and non-invasive early-stage non-small cell lung cancer nodules. Thus, DNA methylation markers hold promise for the development of non-invasive diagnostic assays for early-stage lung cancer.

However, the clinical implementation of DNA methylation markers requires addressing several challenges related to the early detection of lung cancer. First, the sensitivity of methylation assays is compromised in the case of pre-invasive lung cancer lesions and nodules smaller than 1 cm. Second, the methylation status of germline DNA in whole-blood specimens reflects tobacco exposure history rather than tumor-specific alterations. The clinical implementation of DNA methylation markers requires their combination with clinical risk stratification tools or imaging to identify positive indeterminate nodules and avoid unnecessary biopsies. The standardization of assay preparation from different types of specimens and the definition of optimal thresholds in appropriately powered prospective cohorts are also critical for the successful translation of methylation markers into the clinical practice.

2. Circulating Tumor Cells (CTCs)

CTCs are biomarkers that have demonstrated high levels of importance in identifying lung cancer. Their potential lies in the fact that they are caused by tumors and are essentially a precursor for metastasis, showing proven results with early-stage lung cancer, differentiating benign and malignant pulmonary nodules, and more. With a large range of applications across the eight studies that were reviewed, combining CTCs with LDCT screening improved sensitivity by 17.63% (from 57.57% to 75.2%) and specificity by 26.2% (from 62.84% to 89.04%), though there was a slight decrease in AUC by 0.042 (from 0.87 to 0.828). Evaluating these results, it supports the idea that in high risk patients, CTCs have the potential to reduce the high rate of false positives often found when only LDCT screening is utilized. Clinically, the higher sensitivity additionally confirms the effectiveness in distinguishing between benign and malignant tumors or nodules early on, allowing for increased action, less testing later on, and more. Furthermore, the differences in methodology, from utilizing CTC assays to FISH, ISEF methods, and experimental technologies like the CellCollector and microfluidic platforms could account for the reduction in AUC. Overall, combining CTCs with LDCT screening has proven itself to be more effective at detecting lung cancer at an early stage than LDCT alone in high risk patients, however moving forward more standardization with CTC capture technologies and evaluation over time needs to be done to validate results.

3. Circulating Tumor DNA (ctDNA)

Circulating tumor DNA (ctDNA) provides strong potential as a minimally invasive biomarker for lung cancer, particularly for postoperative minimal residual disease (MRD) detection and recurrence prediction. It provides a reasonable efficacy across a broader range of applications than early-stage diagnosis for lung cancer. Quantitative analysis of the included studies indicated that ctDNA had an average sensitivity of 62.8% and a remarkable average specificity of 98.1%. The average area under the curve (AUC) value is also 0.729, presenting a high discriminative power between high-risk lung cancer patients with healthy control groups. Detection of ctDNA after curative-intent treatment was frequently associated with an increased risk of recurrence, indicating that ctDNA may be clinically significant as a biomarker. However, the limited evidence directly comparing ctDNA with LDCT for primary screening does not guarantee an immediate replacement for imaging.

As of now, the findings suggest that ctDNA may not have the full capacity to immediately replace LDCT screening for lung cancer in high-risk patients. The best solution forward, instead, would be to incorporate a multimodal strategy, where the ctDNA biomarkers are combined with the imaging technique to maximize the diagnostic performance. Further research should prioritize large, prospective, multicentre studies and directly compare ctDNA against LDCT. They should also prioritize their focus on early-stage detection than recurrence as many studies have already provided evidence for that. Thus, a minimally invasive and highly accurate diagnosis can be made for lung cancer in the future of medicine.

4. Protein Biomarkers (ProGRP)

Protein biomarkers, in addition to LDCT for high-risk patients, show a strong potential for future research. Across the eight studies, the approaches had an average sensitivity of 88.53% compared to the 57.57% sensitivity for LDCT alone. Protein panels should be used with LDCT rather than replacing it. It should be used before or after imaging, and used to help identify high-risk patients, distinguish malignant from benign nodules, and reduce unnecessary biopsies. Combining panels with CT findings, nodule size, smoking history, and risk calculators can provide the most useful risk assessment in the future.

Before routine use, panels require validation in larger, diverse high-risk populations, including patients with benign lung disease, COPD, and smoking-related inflammation. Future research should identify panels that can be converted into affordable, standardized assays with rapid turnaround times and integrated into carefully designed LDCT screening pathways.

5. Autoantibodies (AABs)

In conclusion, circulating AABs may have some clinical value as ancillary biomarkers in the risk assessment for developing lung cancer, especially in the management of uncertain pulmonary nodules. While most studies found a high degree of specificity, the generally modest and sometimes highly variable sensitivity suggest that AABs are unsuitable as a single screening, diagnostic or negative test. It is most likely that their use is of the most benefit in conjunction with standard screening and diagnostic methods such as computed tomography and the use of

clinical risk factors. Using AAb tests in conjunction with these other factors may further refine risk stratification, helping clinician determine if a nodule may possibly have malignancy, thus aiding decisions on imaging surveillance, biopsy, or further investigation, and possibly obviating the need for potentially invasive procedures.

Further work needs to ascertain how testing for AAb levels can be utilized in the best way with a view to establishing the place and method of testing in the clinic. The marked variance found between studies as to the diagnostic performance of AAb levels, points out the importance of uniform assays, definition of study groups, and consistent clinical outcomes and objectives. Prospective studies conducted in multiple centers with robust designs of large patient numbers followed for a period of time should be undertaken to confirm reproducibility, economic and clinical value of AAbs, in conjunction with other measures in uncertain pulmonary nodules. In the end, testing for AAbs will probably not supersede conventional tests but will be an addition to the panel used in individual risk assessment.

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